

Microbiology 101

Teaching dossier

Cilicia Nascimento

2025

Contents

My teaching philosophy	3
Course syllabus	4
Who will be teaching?	4
What is this course about?	4
Grading	5
Calendar	7
Resources	13
Lesson Plan	14
Small Group activity	17
Assessment	18

My teaching philosophy

As an educator, I am deeply committed to fostering an inclusive, engaging, and dynamic learning environment where all students feel valued, respected, and empowered. My classroom is a space where diversity is celebrated, and every student—regardless of racial, religious, ethnic, gender, or socioeconomic background—is welcomed and encouraged to bring their full self into the learning experience. I strive to create an environment where students feel safe to express their thoughts, explore new ideas, and challenge their perspectives in a respectful and supportive atmosphere.

I believe that learning is a lifelong journey, not a destination. My passion for both teaching and learning drives me to continually refine my methods to ensure that I am meeting the diverse needs of my students. Education should ignite curiosity, inspire critical thinking, and encourage discovery. Therefore, I encourage open dialogue, active participation, and collaboration, recognizing that each student brings unique experiences and perspectives that enrich our collective understanding.

Recognizing that no two students learn in the exact same way, I take a flexible and student-centered approach to teaching. I employ multiple teaching strategies—such as experiential learning, project-based assignments, discussions, and technology-enhanced instruction—to support various learning styles and needs. My goal is to provide students with the tools, resources, and encouragement they need to take ownership of their learning and apply knowledge in meaningful ways.

Science is collaborative by nature, and I encourage teamwork and communication within the classroom. I create opportunities for students to work together on experiments, research projects, and discussions. By fostering an environment where students feel comfortable engaging with one another and with me, I aim to build a supportive community of learners who challenge and inspire each other.

I welcome feedback and see teaching as an evolving practice that grows with the needs of my students. I invite students to engage in shaping the course by suggesting topics, discussions, or learning methods that resonate with them. This collaborative approach ensures that the learning experience remains relevant, engaging, and effective for everyone.

I want my students to leave my class not only with a deep understanding of microbiology but also with a lasting appreciation for the role of science in shaping our world. It's crucial to me that students know that their voices matter, their perspectives are important, and their potential is limitless. By cultivating a classroom culture of respect, curiosity, and inclusivity, I hope to inspire a passion for lifelong learning that extends beyond the classroom and into every aspect of their lives.

Course syllabus

Classroom: - / Laboratory: -

Who will be teaching?

My name is Cilicia Nascimento. I come from the beautiful country of Brazil, I was born in our largest city, Sao Paulo, so I am used to a chaotic life. When I was in undergrad and took an introductory microbiology course, I did not know I would eventually get a PhD in microbiology and come to America. During that course I had this feeling of “this is awesome, I need to know more!” It is my hope that, by the end of our course, you will share that same feeling. One of my favorite activities outside of the lab/classroom is crafting, and crochet is my current passion: I am proud to show you my latest creation. Having a hobby is very important and I hope we have time to share what our favorites are!

Email: cilicia.nascimento@bcm.edu

Phone: (713) 307-2572

What is this course about?

In this course we will explore the micro-world. Be ready to answer important questions, such as: What are bacteria? Do we really have more microorganisms in our body than human cells? What is the difference between bread mold and mushrooms? What is a pathogen? Why do we get food poisoning?

We will have a very inclusive approach, so there is no need to worry about having a science background. But I believe that understanding scientific literature is an important skill, so we will learn how to approach a scientific paper and discuss relevant articles at the end of every unit (check “Unit Science time!” on our calendar below). By the end of this course, microbiology will no longer be a mystery, and you will be ready to join scientific discussions about important microbial topics in today’s world.

By the end of this course, you should be able to:

- Apply multiple techniques, including both microscopy and biochemical tests, to identify several microorganisms, and provide rationale for every reaction or test
- Understand how microbiology concepts apply to our daily lives (for example, why is it important to thoroughly wash our hands but we also tell children to not be afraid to play in the dirt?)
- Engage in discussions about scientific literature on the topic of microbiology (why is it important to share about bacteria resistant to multiple antibiotics?)
- Know the importance of the human microbiome and apply that knowledge to your routine
- Interpret microbiology-related news
- Attend and succeed in advanced microbiology classes

How do we achieve those goals?

- Following the lectures and taking full advantage of the accompanying textbooks and supplemental materials will ensure optimal comprehension of all the content covered in this course
- We will have several activities that will assess your knowledge regarding what was covered in class, so completing them is essential for the development of your understanding

- Have a question? Always ask! This is a learning environment, so all questions are welcome, both during class and during my office hours
- I will give feedback on all activities to help guide your microbiology-learning journey
- A teacher's assistant will be present in class during our reviews at the end of each unit, they will also be available during office hours, so feel free to take your questions to them

Our learning resources:

Main textbook: "Microbiology: an introduction"; Tortora, Gerard J., Berdell R. Funke, and Christine L. Case. Published by Pearson. 13th or 14th ed.

Auxiliary textbook: "Brock Biology of Microorganisms"; Madigan, Michael T., Bender, Kelly S., Buckley, Daniel H., Sattley, Mathew W., Stahl, David A. Published by Pearson. 15th or 16th ed.

Grading

Your grade will have three components:

- **Class activities:** we will have one presentation and one activity that should be performed in groups. For these activities, groups will grade each other. I will evaluate the grades and attribute a grade according to your feedback.
- **Quizzes:** at the end of every unit, we will have a quiz that will be taken online, due one week after the last class of that unit.
- **Final assignments**
 - **Content assignment:** this assignment must show what you learned throughout this course. The form this assignment will take is chosen by each student. Some options are essay, presentation, video, lecture, quiz and oral quiz. All final assignments are due on the day appointed in the calendar below. In the case of presentations or lectures, those can be given in class on that day.
 - **Self-assessment:** one week before the due date of the final assignment, I will have the "self-assessment" form available to be filled out online. The form will cover how this course impacted your knowledge on the subject, and how it will affect your career moving forward.

IMPORTANT: The weight of each grade component will be determined by each student.

Weight options are:

- a) 50-20-30
- b) 60-20-20
- c) 50-25-25
- d) 40-30-30

For example, you chose option c), and decided to distribute the weight as follows:

Class activities: 50%

Quizzes: 25%

Final assignments: 25%

So that should be the components and weight of your final grade.

Due dates

All activities are due one week after their assignment. The dates are marked in the calendar, and I'll communicate any changes in advance. The knowledge acquired through such activities is essential for a better understanding of the following lectures, so their non-completion or delay can create gaps in knowledge. However, I will be accepting quizzes with no grade deduction up to a week after the due date. I cannot accept submissions any later than that to avoid accumulating a large volume of corrections by the end of the semester. For the in-class activities, please let me know if you won't be able to attend or if a reschedule is needed, I will try my best to adjust accordingly.

About "Unit Science time!"

Science is not static; it is constantly evolving. One way that we scientists keep up with what is happening in labs all over the world is through publications in scientific journals. In this course we will not only learn the history of microbiology and the important discoveries of the past, but we will also keep up with the latest research. We will do that through the "Unit science time!", which happens at the end of every unit. I will choose one (or more) recent scientific publications for us to read, understand, and discuss in class. What experiments were done? Are the conclusions coherent with the results that were obtained? Reading, discussing and understanding scientific publications is a valuable skill that will help you in your academic and professional journey, so this will be a great opportunity to develop that skill.

Calendar

Date	Class title	Class summary	Assignments
Unit 1 – A microbe introduction: Here we get familiarized to the concept of “microbiology” and learn about important discoveries that were made in the field throughout the years.			
	What is microbiology? - A brief history of germs	We will talk about the discovery of microorganisms and draw a timeline from the germ theory until today.	
	Under the microscope	We will understand the principles and components of the microscope, a brief history of microscopy. I will go over types of microscopes and we will learn about several different stains. This lecture includes a laboratory activity.	
	Building bacteria	We will learn about the structure and function of several different bacterial components, like cell walls, membrane, flagella, and others.	
	HOLIDAY – Labor Day		
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Quiz1: to be taken home I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
Unit 2 – Bacterial functions and identification: Here we will learn how bacteria have been surviving and thriving in different environments and how we can identify a bacterial species based on what we know about them			
	Bacterial reproduction and genetics	We will learn about bacterial replication, transcription and translation. We will	

		also talk about regulation of expression and recombination.	
	Bacterial metabolism	We will learn how bacteria obtain energy; we will participate in an activity that illustrates the electron chain. And I will show how bacteria metabolize complex molecules.	Quiz1: DUE DATE
	DNA technology	We will go over the principles of different molecular biology techniques (like PCR, sequencing, cloning and others).	Optional: bring a sample you'd like to see in a petri dish
	Bacterial identification	This lecture will be in the lab . We will perform different biochemical tests and learn what is the principle behind each one.	A step-by-step report on how to identify a bacterium from the sample analyzed by your group. You should include what microorganism(s) is(are) present according to your results. And what molecular biology techniques could be used to confirm your results.
	Molecular biology techniques	This lecture will be in the lab . We will perform a PCR targeting specific bacterial genes and run an electrophoresis gel to visualize our results	
	Pathogenicity	We will learn about mechanisms used by microorganisms to establish colonization and infection	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a	Lab Report: DUE DATE Quiz2: to be taken home.

		closer look at concepts that might still be cloudy.	I will upload the articles that will be discussed in the next class
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
Unit 3 – Learning about specific microorganisms Here we will learn the names and last names (genus and species) of important bacteria, and how they relate to human health and disease			
	Gram-negative bacteria	We learned about Gram staining, now we will go over the main microorganisms that belong to the Gram-negative group.	In groups : each group will choose one Gram positive bacteria and develop an activity based on it. That should include what we learned about bacteria so far.
	Gram-positive bacteria	This lecture will be a collaboration between students. Each group will apply their activity to the whole class (including myself).	Quiz2: DUE DATE Grading each other's Gram-positive activity.
	Microbial diseases	Let's learn about some bacteria of medical importance. We will discuss diagnosis, symptoms and treatment. We will also talk about antimicrobial drugs.	We will talk about the assignment where you will prepare a lecture about two microorganisms of your choice (from a list). That lecture will be given in the next unit.
	The human microbiome, what we know so far	We will learn about the Human microbiome project, main characteristics of microbiomes from several sites in the body. And how our knowledge about microbiomes is shaping modern medicine.	
	Epidemiology	We will understand how disease spreads and how it affects society.	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all	Quiz3: to be taken home.

		questions and take a closer look at concepts that might still be cloudy.	I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
	HOLIDAY: Columbus Day		
Unit 4 – Time to apply our knowledge! In this unit we will dive into specific microbial infections.			
	Microbial infections in humans: skin and eyes	During these, each lecture will be divided into 2 parts: in the first one I'll introduce the subject and talk about many examples of microbial diseases pertinent to the class, in the second part, one or more groups will give a presentation about one microorganism previously chosen.	Group presentation
	Microbial infections in humans: skin and eyes		Quiz3: DUE DATE Group presentation
	Microbial infections in humans: the cardiovascular and lymphatic systems		Group presentation
	Microbial infections in humans: the respiratory system	During these, each lecture will be divided into 2 parts: in the first one I'll introduce the subject and talk about many examples of microbial diseases pertinent to the class, in the second part, one or more groups will give a presentation about one microorganism previously chosen.	
	Microbial infections in humans: the digestive system		Group presentation
	Microbial infections in humans: the urinary and reproductive systems		Group presentation
	Environmental microbiology		Group presentation
	Unit review	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Grading each other's lectures. I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles	

		that are pertinent to this unit to be discussed in class	
Unit 5 – Broadening our horizons We have explored the amazing microbial world with a special emphasis on bacterial human infections, now it is time to learn about viruses and fungi			
	Introducing virology	We will discuss the main human viruses' infection mechanisms and discuss about what changed in our epidemiological approach since the SARS-CoV-2 pandemic.	
	Virus structure and replication	We will learn the mechanisms involved in viral reproduction	
	Human viral infections	We will discuss the most important viral infections and build a timeline of viral pandemics throughout history.	
	Environmental virology	We will learn about the viruses that inhabit our environment, and how humans have adapted to them and vice versa.	
	Introduction to Fungi	We will talk about the kingdom Fungi, that encompasses a multitude of organisms.	
	Fungi of medical importance	We will discuss a few fungal infections, including their symptoms and treatment.	
	Fungal Metabolism	We will learn the unique mechanisms involved in how fungi obtain energy and thrive in different environments.	
	Environmental importance of Fungi	We will go over how viruses can influence the environment and the latest biotechnological	

		advances in fungal science.	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Quiz4: to be taken home. I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
	HOLIDAY - Thanksgiving		
	Mega review	We will go over the burning questions regarding all topics we went through this semester. We will have a review activity that does not account for grading. And I will explain what the final assignment will look like and what I expect from it.	Quiz4: DUE DATE
	Final assignment	Each student will take their final assignment as they chose previously. The final self-assessment will be filled out online.	
	Last day of class!	I will make your final grade available for you one day before our last day of class. And I will be available in class during our last day to discuss grades and answer questions.	

Resources

Blackboard

https://bcm.blackboard.com/ultra/courses/_5394_1/cl/outline

Lesson Plan

Teacher: Cilicia Nascimento	Subject/Audience: Bacterial reproduction and genetics / undergraduate students (freshman, sophomore)	Date: 09-08-2025
STEM+M Classroom Strategy Focus: <i>Which STEM+M strategies will be embedded in the lesson?</i> (Refer to Active Learning Database) • Mini quiz • Concept Mapping • Short Cases/Scenarios		Learning Objectives: We will learn about bacterial replication, transcription and translation. We will also talk about regulation of expression and recombination (in bacteria).
Metacognition: <i>How will learners process their thinking?</i> By watching bacterial genetics mechanics in videos and then role-play the concepts and recreating what we learned, students will process concepts from multiple points of view and build their knowledge by applying it to the activities.		Real-World Connections: <i>How does this lesson connect to the daily lives of students?</i> Bacterial reproduction is happening all the time both in the environment and in our bodies, so understanding it will help students connect all future information about bacteria to the basic concepts taught in this lesson.
Standards: <i>What standards will be covered in the lesson?</i> • Understand how microbiology concepts apply to our daily lives Attend and succeed in advanced microbiology classes • Engage in discussions about scientific literature on the topic of microbiology		
Essential Questions: <i>What essential questions will be covered in the lesson?</i> • What is molecular biology's central dogma? • What is replication, transcription and translation? • What is the path from DNA to protein? What mechanisms are involved? • What is DNA mutation? What are the main types of mutation? • How does bacteria repair DNA?		
Differentiation Strategies: <i>How will the lesson meet the needs of diverse learners?</i> We will show animations of all processes covered in this lesson, so that the concepts are more easily visualized. In addition to that, we will have pair and group activities, including role-playing, so all students can participate and contribute to this class.		

Materials: <i>What materials are needed?</i> We will use crafting materials provided by me and students will use their cellphone for some activities.	Technology Integration: <i>What technology tools will be used to support learning?</i> • Videos (animations) of molecules involved in bacterial replication, transcription and translation • We will access previous knowledge using Poll Everywhere • We will use Mentimeter to learn what words students associate with “mutation”
Engage: How <i>will I capture learner interest and establish prior knowledge?</i> Before starting today’s lecture, I will pose a few key questions and access the classroom understanding of previous knowledge using Poll everywhere. This way the answers are anonymous, so students don’t have any reason not to participate. And I can access what concepts I need to reinforce throughout the class. Example questions: Do bacteria have a nucleus? Options: Y, N. Which of these is a purine base? Options: A, T, C Which of these is directly connected to DNA replication? Options: Ribosomes, Polymerases, Mitochondria	
Explore: How <i>will learners construct knowledge through questioning and actively engage in a learning task?</i> After a brief exposition of the concepts of Replication, Transcription and Translation, I will provide building blocks representing DNA and RNA nucleotides, amino acids, and a long double-helix DNA strand. Students will roleplay as the enzymes involved in DNA replication (DNA polymerase, helicase, etc.) and will need to build two new DNA strands from the one that was provided. Always connecting with the appropriate number of hydrogen bonds. Afterwards, another group will be representing the enzymes involved in transcription and translation and will use the building blocks to construct mRNA and a short peptide chain. Students need to find the promoter region and provide the correct codon-anticodon combinations. I will provide large nametags for the students roleplaying as enzymes. This activity should involve the whole class, even if only a limited number of students is actively involved with the building blocks, the class should provide guidance for each step of the process.	
Explain: How <i>will learners interpret, clarify, and refine learning?</i>	

I will provide an exposition about different types of mutations found in bacteria, then the classroom will be divided into groups, and each one will receive one type of mutation (silent, missense, point mutation, etc). And they must explain how that mutation happens using provided materials (small building blocks, DNA strands). Drawing on the board is also possible.

Elaborate: *How will learners apply and generalize concepts?*

Evaluate: *How will I assess learning?*

In pairs, students will receive a short excerpt of a scientific article describing a specific mutation in a microorganism that caused changes in virulence or antibiotic resistance.

In class, students will provide a description of how the processes of transcription and translation are affected by comparing what happens in the wild type to the mutant. Students will submit short essays, and we will discuss them in class. There will be repeated examples, since the class is large and would not have time to discuss the activity otherwise.

Timeline

10:30am – Greetings, Poll Everywhere activity, reinforcement of key concepts that might not be totally clear.

10:40am – Exposition time about replication, transcription and translation. Including video animations.

11:00am – Role playing activity

11:20am – Listening to the students' experience, covering any doubts that came up during the activity.

11:30am – Mutations Mentimeter, save word bank for further analysis.

11:35am – Exposition time about DNA damage and mutations

11:50am – Group activity about mutations

12:00am – Feedback on activity

12:05am – Short scientific article activity

12:20am – Short feedback and discussion

12:30am – End of class

Small Group activity

Group investigation

Students will be divided into small groups; each assigned a specific type of DNA mutation and its underlying mechanisms from a provided list. I will assign the groups, covering topics such as silent, missense, and nonsense mutations, as well as spontaneous vs. induced mutations and classes of mutagens.

Each group will have 15 minutes to:

- Research the definition, mechanisms, and examples of their assigned topic.
- Write a brief summary of what they learned.
- Prepare a short (3-minute) presentation using provided materials to illustrate how the mutation occurs.

Afterward, groups will pair up to present their findings to each other. Each interaction will last approximately 6 minutes, with additional time for questions and discussion. The entire activity will take about 20–25 minutes, encouraging students to grasp the topic and summarize key points concisely.

Since all groups will both present and observe a presentation, each group will receive a rubric with the following reflection questions to provide feedback:

- Do I have a better understanding of the topic after the presentation?
- Can I explain the concept based on the information presented?
- Was the presentation clear and well-structured?

I will collect the feedback, but this activity will not be graded.

Assessment

Using a Google Forms students will answer three questions from this lecture

1. Three things you learned in the lecture
2. Two things you found particularly interesting
3. One question you still have from what we covered

I will evaluate the answers and assess if the students have a satisfactory understanding of the content and will collect all the information from the third question and cover the main topics in the next lecture.